Week Summary

Memory and cache

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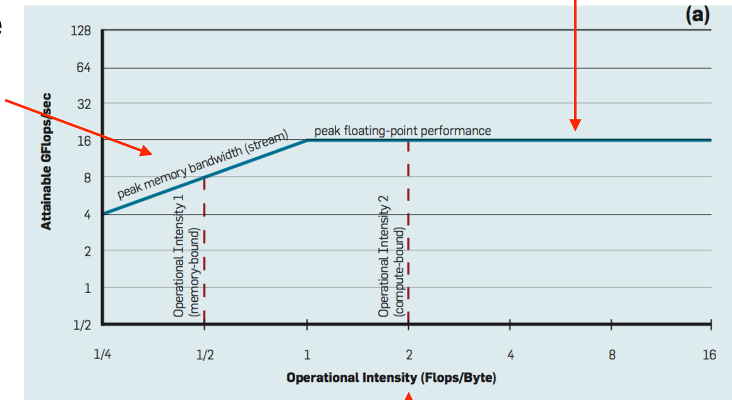
Last week we saw how to calculate the theoretical maximum performance (R_{peak}) which assumes that floating point hardware is working with 100% efficiency

In practice performance may be limited by the rate at which data can transferred to and from memory
(memory bandwidth)



An application may be compute bound or memory bandwidth bound depending on the arithmetic intensity of the algorithms used

Summary

- By considering arithmetic intensity we can understand how algorithms influence performance at the level of a single compute node
- The implementation of the algorithm is also important e.g. making effective use of cache
- In HPC we need to consider how a parallel program maps onto the hardware e.g. NUMA effects